

Clinical Early Warning System: Deep Learning for Sepsis Detection

Department of Artificial Intelligence — Deep Learning Assignment Report
PhysioNet Sepsis Prediction Dataset | 546,123 Records | 44 Features | 2.17% Sepsis Rate

Section 1: The Clinical Problem

Patient deterioration prediction differs fundamentally from standard classification tasks. In image or text classification, each sample is independent. In the ICU, a patient's vital signs form a trajectory across hours and days — a single low blood pressure reading may be a measurement artefact, but a declining trend over six hours is a medical emergency. The data is irregularly sampled, riddled with missing values, and carries severe class imbalance: only 2.17% of the 546,123 records in this dataset represent sepsis onset. A naive classifier that always predicts no sepsis would achieve over 97% accuracy while being clinically worthless.

Tabular vital signs alone capture the what but not the why or the context. A heart rate of 115 bpm means something different in a post-operative patient versus a patient with no recent intervention. Free-text clinical notes written by nurses and physicians encode this context: observations about mental status, skin colour, patient-reported symptoms, and clinical reasoning that cannot be reduced to a number. A note reading 'patient confused and diaphoretic despite adequate BP' signals sepsis risk in a way that structured vitals cannot. The best early warning systems must read both.

Recall matters more than accuracy here. Consider a concrete scenario: a 72-year-old post-surgical patient, Mrs. A, shows borderline vital signs at hour 6. The model flags her — a false positive. The cost is one additional nursing assessment. Now consider the alternative: the model misses her. By hour 10 she is in septic shock, requiring vasopressors and ICU transfer. The Surviving Sepsis Campaign estimates each hour of delayed treatment increases mortality risk by approximately 7%. A missed case is potentially fatal; a false alarm is a minor inconvenience. This asymmetry demands that recall, not accuracy, be the primary optimization target in any life-critical early warning system.

Section 2: Baseline Design Decisions

The DNN baseline used a three-layer architecture: 256 neurons with ReLU, Batch Normalisation and Dropout (0.4); 128 neurons with ReLU, Batch Normalisation and Dropout (0.3); and 64 neurons with ReLU, feeding a sigmoid output. Two versions were trained on 20,000 subsampled records balanced with SMOTE, one with SGD ($\text{lr}=0.001$, $\text{momentum}=0.9$) and one with Adam ($\text{lr}=0.0001$).

The vanishing gradient problem arises because backpropagation multiplies gradients through repeated weight matrices and activation derivatives as it travels toward the input layers. In networks with many layers, these products become exponentially small, preventing early layers from learning. Two choices countered this. First, ReLU activation: its derivative is exactly 1 for positive inputs, so it does not shrink the gradient. Sigmoid and tanh saturate in their tails, compounding the problem. Second, Batch Normalisation normalises each layer's input to zero mean and unit variance, preventing activations from drifting into saturated regions. It is also worth distinguishing loss from cost: a loss function measures error on a single example (here, binary cross-entropy on one patient record), while the cost function aggregates loss across the full batch. The optimiser minimises cost, which is why validation cost is tracked separately — a model can memorise individual examples while generalising poorly.

SGD with momentum substantially outperformed Adam on this dataset. SGD converged more slowly but reached a superior solution, while Adam's adaptive learning rates may have stepped too aggressively given the high-dimensional, SMOTE-augmented feature space. The results below confirm this.

Model	Accuracy	Precision	Recall	F1	Train Time (s)
DNN + SGD	0.8900	0.8330	0.9774	0.8995	64.8
DNN + Adam	0.7422	0.7522	0.7278	0.7398	24.5

Dropout contributed meaningfully to generalisation. By randomly deactivating 40% of neurons per forward pass during training, it prevents co-adaptation: no neuron can rely on the presence of specific others, forcing the network to build redundant representations. At inference all neurons are active, their outputs scaled by the survival probability, approximating an ensemble of smaller networks. The training-validation accuracy gap was 4.1% — well below the 7–10% gap typical of unregularised networks of this depth, confirming that regularisation was effective.

Section 3: Modelling Patient Timelines

The DNN treated each hourly record as an independent observation, discarding the temporal structure that defines clinical deterioration. Generation 2 restructured the data into sliding windows of 12 consecutive hours (yielding 1,068,534 sequences of shape 12×48) and trained three recurrent architectures on the full SMOTE-balanced dataset.

Standard RNNs suffer a severe form of the vanishing gradient problem. During backpropagation through time, gradients are multiplied by the same recurrent weight matrix at every time step. Over 12 steps, if the matrix eigenvalues are less than 1, the gradient shrinks exponentially — effectively erasing any signal from early time steps. An RNN cannot learn clinical patterns that span more than three or four hours, which is clinically insufficient. LSTM solves this through a dedicated cell state that updates by addition rather than multiplication. The new cell state is simply a weighted blend of the old cell state and new candidate information, controlled by two learned gates. The gradient along this pathway is multiplied only by the forget gate value, which is a number between 0 and 1, rather than an entire weight matrix. When the forget gate learns to stay near 1 for important clinical information, gradients travel across all 12 time steps unimpeded. This explains LSTM's highest recall of 98.8% and its stable loss curves through 15 epochs.

GRU achieves a deliberate simplification: it merges the cell and hidden states and reduces three gates to two (update and reset). What it gives up is the separate memory pathway — it cannot independently gate what to store from what to output. For very long sequences or tasks needing precise long-range memory control, this is a real limitation. What it gains is speed: 701.7 seconds versus 2,103.6 seconds for LSTM, a 3.0x reduction. For a clinical system requiring frequent retraining on incoming hospital data, this is practically decisive.

Model	Accuracy	Precision	Recall	F1	Train Time (s)
LSTM	0.9924	0.9968	0.9879	0.9923	2103.6
GRU	0.9885	0.9999	0.9771	0.9884	701.7
BiLSTM	0.9886	0.9997	0.9776	0.9885	887.1

For real-time ICU monitoring, unidirectional models are the only valid choice. A live system makes a prediction at the current hour using only what has already happened. Bidirectional LSTM processes sequences in both directions, meaning it requires future vital signs — values that do not yet exist at prediction time. Using BiLSTM in a live system would require waiting until the patient's full admission history is available, defeating the purpose of early warning. Its appropriate use is retrospective: post-hoc research audits, mortality prediction on completed stays, or training data analysis where the full trajectory exists. GRU is therefore the recommended architecture for deployment, combining near-LSTM recall with a 3x speed advantage and unidirectionality. Its recall of 0.977 also compares favourably with the qSOFA score, which reported

sensitivity in the range of 0.29–0.60 in the Sepsis-3 validation study — a substantial gap that reflects GRU's ability to model deterioration trajectories rather than a single-point bedside assessment.

Section 4: The Transformer and Clinical Language

ClinicalBERT (emilyalsentzer/Bio_ClinicalBERT) is pretrained on MIMIC-III clinical notes and PubMed abstracts. Transfer learning means the model arrives at the task having already learned biomedical vocabulary, clinical abbreviations, the structure of nursing assessments, and semantic relationships between terms like 'hypotension', 'vasopressors', and 'septic shock'. Training a transformer from scratch on 2,000 records would be futile — the model would overfit before learning anything meaningful. Pretraining on millions of clinical documents provides representations that already encode the clinical world; fine-tuning adapts them to the specific prediction task.

The self-attention mechanism is why transformers excel at clinical text. Unlike RNNs that process words sequentially, self-attention computes relationships between every pair of tokens in a single matrix operation. A note reading 'patient confused and diaphoretic despite adequate BP' allows the model to directly connect 'confused' and 'diaphoretic' to sepsis risk regardless of their position, mirroring how a clinician scans a note for salient terms rather than reading left-to-right. Because the transformer has no recurrence, it has no inherent sense of order. Positional encodings — sinusoidal vectors added to each token's embedding — inject this information. This matters clinically: 'blood pressure falling' and 'falling blood pressure' have the same words but different structural implications; 'stable then deteriorated' versus 'deteriorated then stable' represent opposite clinical trajectories. Without positional encoding, these distinctions vanish. The Transformer also offers a practical scalability advantage: its self-attention computes all token relationships in parallel matrix multiplications, allowing a hospital to batch 64 clinical notes and process them in the same wall-clock time as one, making cohort-wide hourly reassessment computationally feasible.

Two fine-tuning strategies were compared on 2,000 balanced records. Frozen BERT fixed all 12 encoder layers and trained only the binary classification head. Full fine-tuning updated all parameters end-to-end.

Strategy	Accuracy	Precision	Recall	F1	Train Time (s)
ClinicalBERT (Frozen)	0.5350	1.0000	0.0700	0.1308	19.8
ClinicalBERT (Fine-tuned)	0.9375	0.8889	1.0000	0.9412	77.2

The frozen model failed catastrophically: 7% recall means 93 of every 100 sepsis patients were missed. Pretrained BERT representations, while rich in language knowledge, are not linearly separable for sepsis classification — the classification head alone lacked the capacity to remap them. Full fine-tuning corrected this, achieving 93.75% accuracy and perfect recall (1.000) in 77 seconds — only 3.9x longer than frozen. The frozen strategy is appropriate when labelled data is very scarce (under ~500 samples) or when only CPU is available; full fine-tuning should be the default whenever 1,000 or more labelled samples and a GPU are available. The perfect recall result must be interpreted cautiously: it was measured on a balanced test set of 200 sepsis cases. A single misclassification shifts recall from 1.000 to 0.995, making this metric highly sensitive at this scale. External validation on a large imbalanced cohort is necessary before deployment conclusions can be drawn.

Attention weight extraction across all 12 layers revealed that the [SEP] boundary token received by far the highest weight (0.5296), with clinically meaningful tokens — vital signs (0.0100), oxygen (0.0094), shock index (0.0072) — accounting for less than 4% of total attention collectively. The [SEP] dominance is a well-documented BERT artefact: attention weights do not reliably indicate feature importance, and models route probability mass to structural delimiters independent of their clinical content. The presence of attention on vital and oxygen tokens is directionally meaningful, but these findings should be treated as exploratory. Gradient-based methods such as Integrated Gradients or SHAP would provide more reliable attribution for

clinical interpretability audits. On the per-class breakdown, the fine-tuned model correctly identified all 200 sepsis cases and 175 of 200 stable patients, generating approximately 1 false alarm per 8 true alerts — a clinically acceptable precision in a supervised workflow.

Section 5: Deployment Verdict

The unified results across all six models are presented below.

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The recommended deployment model for a live ICU early warning system is the GRU. It achieves 97.7% recall and near-perfect precision (99.99%), processes a 12-hour vital sign window in under 100 milliseconds, and retrains in approximately 12 minutes — fast enough to update daily on incoming hospital data. LSTM achieves marginally higher recall (98.8%) but requires three times the training time and provides no latency advantage at inference. BiLSTM requires future data and cannot be used in real-time. ClinicalBERT fine-tuned achieves perfect recall on its test set but was trained on a 2,000-record subset; its recall figure requires external validation and its inference is slower and less interpretable to bedside clinicians than a structured vital-sign model. The GRU's near-zero false alarm rate is also critical: in an ICU where alarm fatigue is a documented patient safety problem, a system that only alerts on genuine risk is far more likely to be trusted and acted upon.

Several ethical concerns must be addressed before deployment. Algorithmic bias is the most serious: the model was trained predominantly on data from specific hospital populations, and performance may differ significantly across demographic groups, comorbidity profiles, or patients from underrepresented ethnic backgrounds. Periodic bias audits stratified by age, sex, and ethnicity are mandatory. Privacy is a second concern — patient vital signs and clinical notes are sensitive data governed by healthcare regulations; the system must operate within a secure, audited infrastructure with strict access controls. Accountability is perhaps the most difficult issue: when the model misses a sepsis case, who is responsible — the clinician who trusted the alert system, or the engineer who deployed it? This question demands that the system be framed explicitly as a decision support tool, never a replacement for clinical judgment. All alerts must be reviewed by a qualified clinician before action is taken, and model limitations must be communicated clearly to frontline staff.

To expand the system to chest X-rays alongside clinical notes, the architecture would need to evolve into a multimodal fusion framework. A Vision Transformer (ViT) would encode X-ray image patches into a sequence of embedding vectors, in the same way ClinicalBERT encodes text tokens. The two streams could be combined through late fusion — averaging or concatenating the final probability scores from each modality — or through cross-attention fusion, where image patch tokens and text tokens interact directly, allowing the model to connect radiological findings with clinical language. For example, a note mentioning 'bilateral infiltrates' could attend to corresponding regions in the X-ray, and vice versa. This architecture would capture pneumonic or pulmonary oedema patterns that frequently precede sepsis but are invisible in tabular vital signs, substantially increasing early detection capability.

Collab Link:

https://colab.research.google.com/drive/1xCRpZvCW_LBDGgqu8961r1UhgyyE7ydR?usp=sharing